Guidewire DEVTrails 2026

SurakshaRoute

AI-Powered Parametric Income Protection for Food Delivery Workers

Protecting weekly earnings when disruption, not injury, keeps gig workers off the road.

Income stops fast. Protection does not.

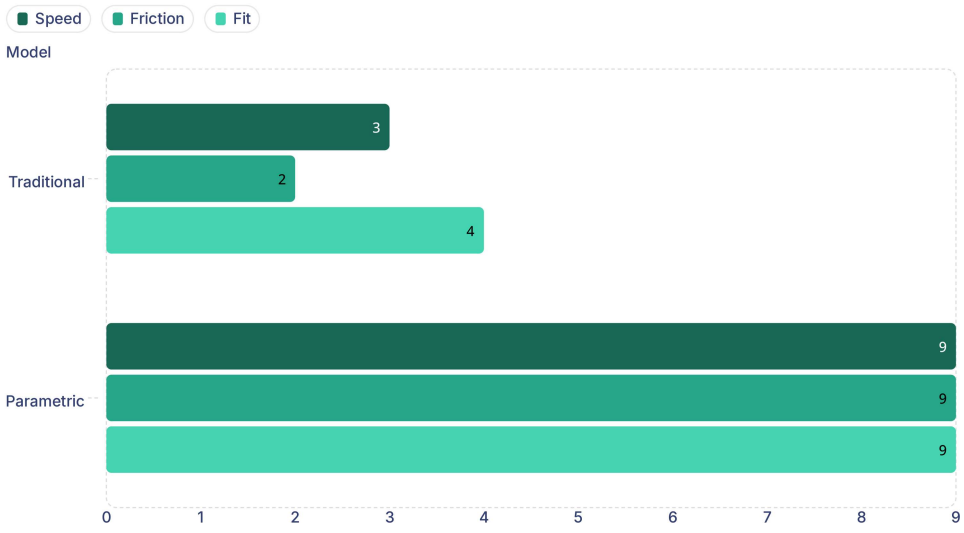
Food delivery workers in India lose income when heavy rain, flood, severe pollution, heatwave, curfew, road closures, or platform outages make work impossible. Traditional insurance is too slow, claim-heavy, and built for loss events, not real-time income interruption. This challenge is strictly about **loss of income** protection.

Why parametric wins

SurakshaRoute is an AI-powered parametric insurance platform for food delivery gig workers. The worker, the weather, and the policy rules connect into one weekly protection loop.

- Worker onboarding
- Risk scoring
- Weekly premium
- Policy activation
- Disruption trigger
- Automatic claim
- Fraud check
- Payout

We chose this persona because food delivery work is high-frequency, time-sensitive, and highly exposed to local disruption.



End-to-end workflow

Worker-side: sign up, activate weekly cover, and receive instant status. Admin-side: monitor triggers, validate fraud signals, and reconcile payouts in one dashboard.

Five AI engines working as one system

1

Risk Scoring

Profiles city-level exposure and worker behaviour to assign a weekly risk band.

2

Weekly Premium

Turns risk, income, and plan choice into a fair weekly price.

3

Fraud Detection

Flags spoofing, suspicious clusters, and inconsistent claim behaviour.

4

Trigger Monitoring

Matches covered disruption events to a worker’s zone and active policy.

5

Payout Estimation

Normalises lost hours and caps payout to keep outcomes fast and consistent.

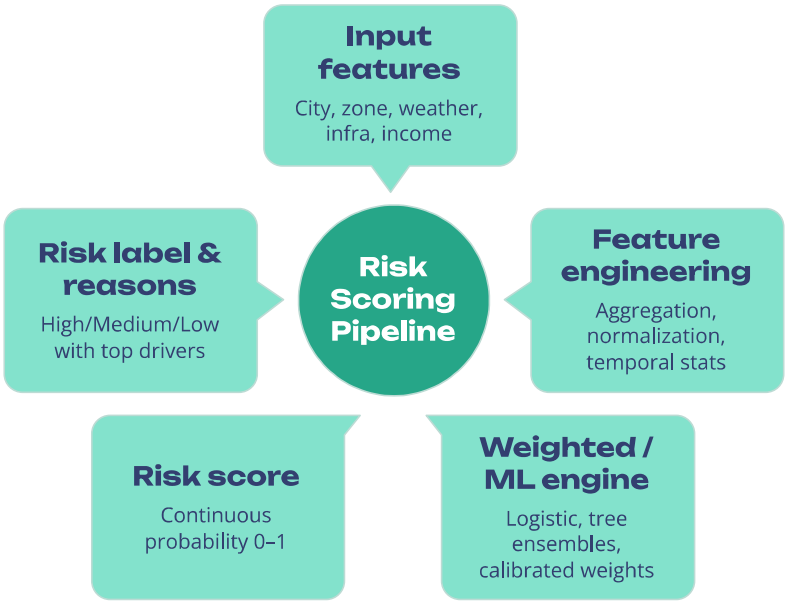
AI/ML architecture I: risk and pricing

Risk Scoring Model

Inputs: city, zone, flood-prone score, rainfall history, AQI history, heatwave frequency, road closure frequency, outage frequency, weekly income, shift timing.

Flow: input features → feature engineering → weighted/ML risk engine → risk score + risk label.

Outputs: score, Low/Medium/High, top reasons.

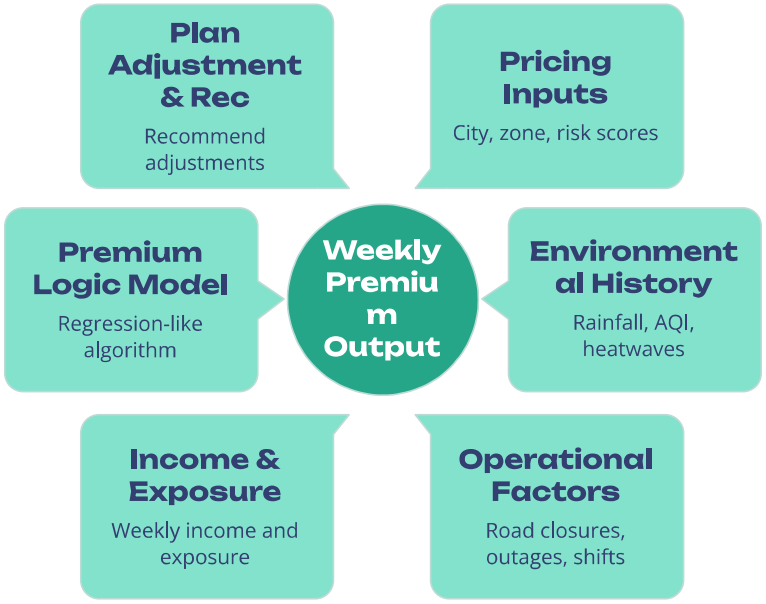


Dynamic Weekly Premium Engine

Inputs: risk score, weekly income, selected plan, disruption exposure, coverage limit.

Flow: pricing inputs → premium logic or regression-like model → plan adjustment → weekly premium output.

Outputs: premium amount, premium breakdown, recommended plan.



AI/ML architecture II: fraud, trigger, payout

Fraud Detection + Anti-spoofing

Inputs: GPS claim, route continuity, shift timing, activity history, zone-event match, device/network consistency, suspicious cluster patterns.

Flow: multi-signal verification → anomaly scoring → spoof detection → fraud score → decision routing.

Outputs: fraud score, level, reasons, approve/hold/review.

Trigger Monitoring Logic

Inputs: weather, AQI, road closure, flood, outage event; worker zone; active policy; shift timing; covered triggers.

Flow: event ingestion → threshold logic → zone match → shift overlap → claim eligibility.

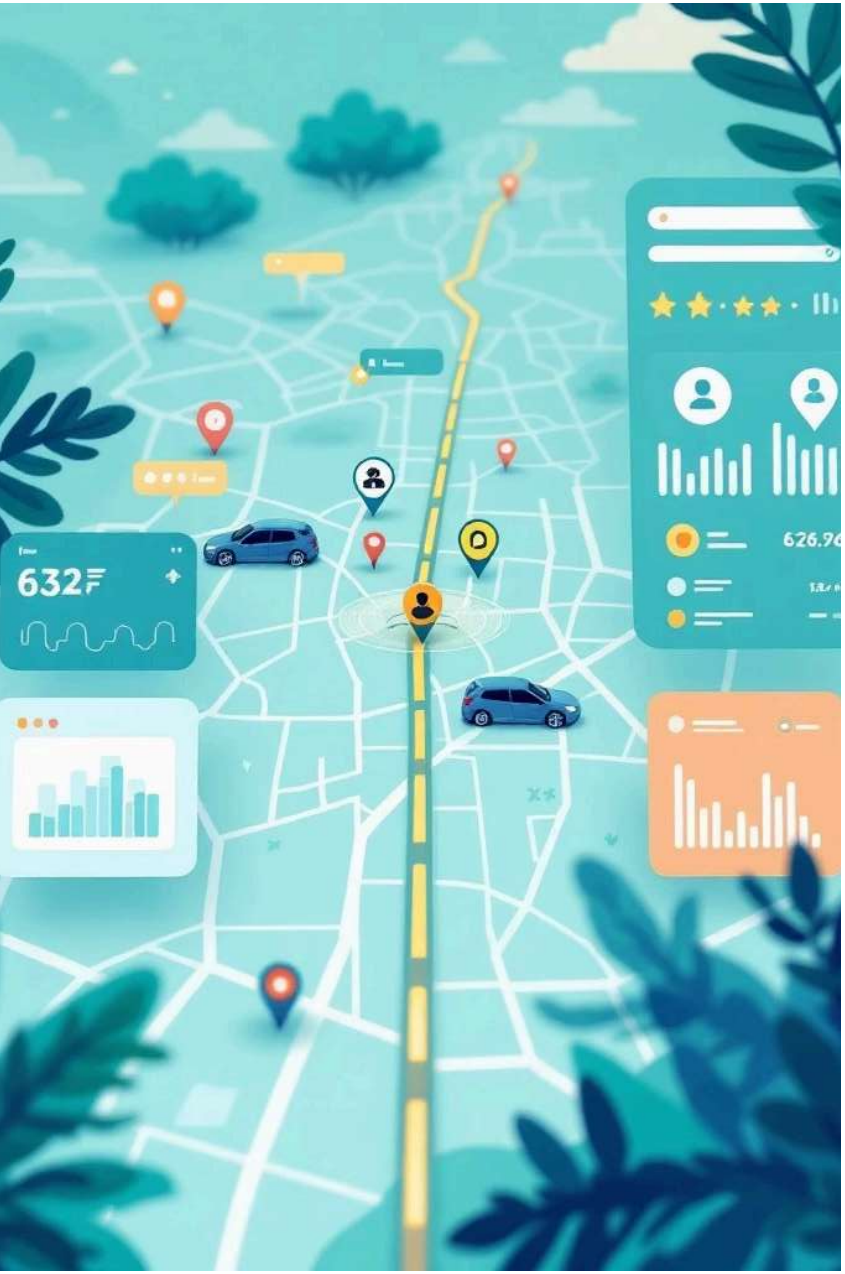
Outputs: trigger matched, claim created.

Payout Estimation Engine

Inputs: weekly income, weekly hours, disrupted hours, coverage %, payout cap.

Flow: income normalisation → disruption loss estimation → cap logic → payout output.

Outputs: payout amount, status, reference ID.

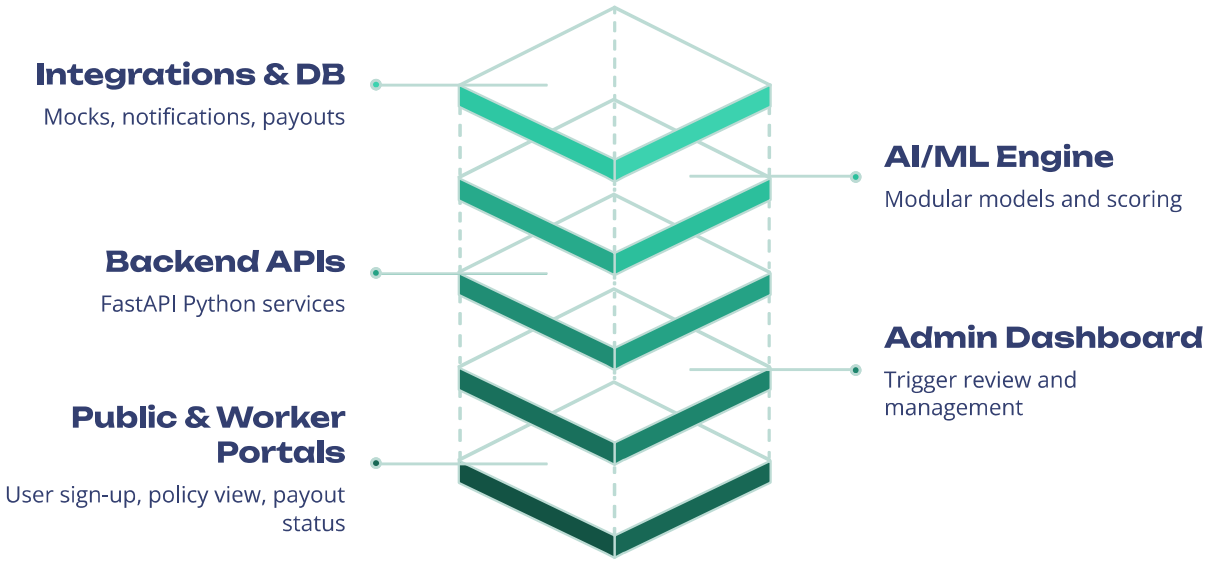


GPS alone is not trusted

We defend against fraud rings that spoof location into severe weather zones. SurakshaRoute cross-checks **GPS + route continuity + activity history + network consistency + event correlation + fraud ring clustering** to distinguish a stranded worker from an attacker.

Fairness rule: low-risk auto-approve, medium-risk soft hold, high-risk manual review. Honest workers are not unfairly rejected because of weak signal or weather-related connectivity loss.

Full system architecture



Stack: frontend dashboard, FastAPI/Python backend, modular AI/ML engines, mocked weather/AQI/traffic/platform APIs, deployed web app.

- Worker portal for sign-up, policy view, and payout status
- Admin dashboard for trigger review, fraud triage, and monitoring
- Backend APIs connecting policy, event, and payout services
- Database and notifications supporting transparent operations

Built for impact, ready to scale

Impact: protects gig-worker income, enables fast zero-touch claims, reduces fraud, improves insurer trust, and scales across cities. The platform can later extend to other delivery personas with the same weekly parametric backbone.

Demo-ready: working AI/ML engines, deployed dashboard, fraud defence, architecture, and workflow.

SurakshaRoute transforms disruption into fast, fair, and intelligent income protection.

Thank you. Q&A.